

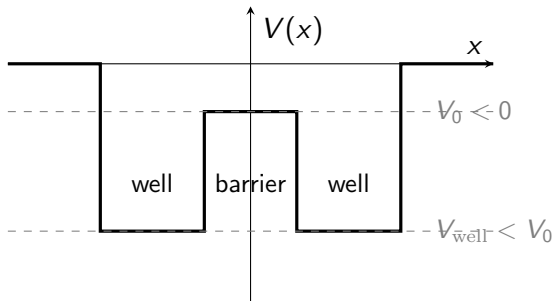
Lecture 07a: bonding

Prof. Daniel Schwalbe-Koda

MAT SCI 120 — Winter 2026

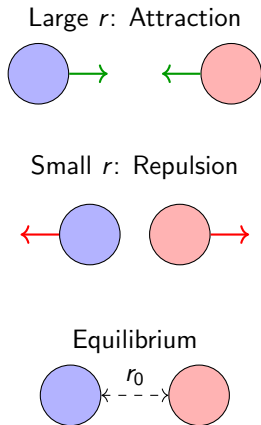
Why do atoms form bonds?

- ▶ Isolated atoms in gas phase have higher total energy than bonded atoms
- ▶ Nature favors configurations that minimize total energy
- ▶ Solids exist because bonded atoms achieve lower energy states



Energy minimization: the driving force for bonding

- ▶ Attractive forces pull atoms together, lowering potential energy
- ▶ Repulsive forces prevent collapse at very short distances
- ▶ Equilibrium occurs where total energy reaches minimum



Bonding energy: quantifying stability

- ▶ Bond energy E_c is the energy released when atoms form a bond
- ▶ Negative E_c indicates stable bonding (energy lowered from isolated state)
- ▶ Typical bond energies: 1–10 eV per atom pair

Condition for bond stability

$$E_c < 0 \quad (1)$$

Notation:

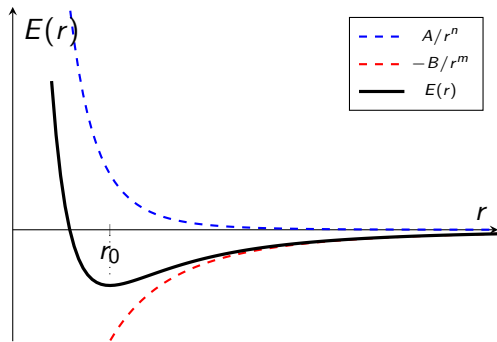
- ▶ E_c — bond energy (cohesive energy)

Universal energy-separation curve $E(r)$

- ▶ Atomic pairs typically exhibit same qualitative $E(r)$ shape
- ▶ Repulsive term A/r^n dominates at small r
- ▶ Attractive term $-B/r^m$ dominates at large r

Energy-Separation Approximation

$$E(r) = \frac{A}{r^n} - \frac{B}{r^m} \quad (2)$$

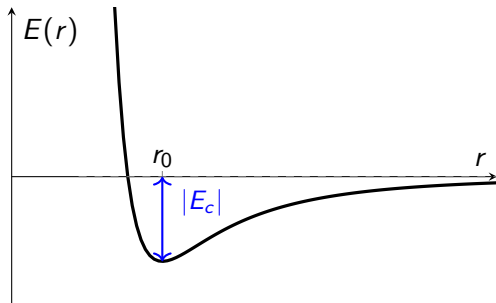


Equilibrium separation r_0 and bond energy

- ▶ Equilibrium: $dE/dr = 0$ (force balance)
- ▶ Solving gives $r_0 = (nA/mB)^{1/(n-m)}$
- ▶ Bond energy E_c is depth of minimum
- ▶ Deeper minimum $\rightarrow$ stronger bond

Bond energy

$$E_c = -\frac{B}{r_0^m} \frac{m}{n-m} \quad (3)$$

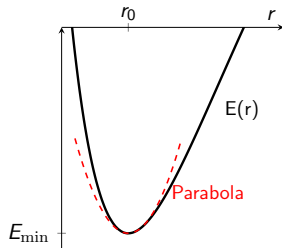


Taylor expansion near equilibrium

- ▶ Expand $E(r)$ about r_0 :

$$E \approx E(r_0) + \frac{1}{2} \left. \frac{d^2 E}{dr^2} \right|_{r_0} (r - r_0)^2$$

- ▶ Linear term vanishes: $dE/dr|_{r_0} = 0$
- ▶ Parabolic approximation valid for small displacements
- ▶ Curvature determines stiffness



Deriving bulk modulus from $E(r)$

- ▶ Under isotropic compression, volume change relates to r change
- ▶ Bulk modulus c measures resistance to uniform compression
- ▶ Factor $1/9r_0$ arises from 3D geometry (volume $\propto r^3$)
- ▶ Larger curvature at $r_0 \rightarrow$ higher bulk modulus

Bulk modulus from atomic interactions

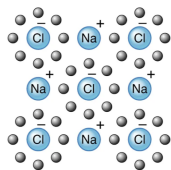
$$c = \frac{1}{9r_0} \left. \frac{\partial^2 E}{\partial r^2} \right|_{r=r_0} \quad (4)$$

Notation:

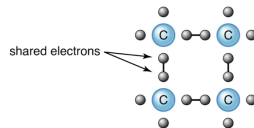
- ▶ c — bulk elastic modulus
- ▶ r_0 — equilibrium interatomic distance

Four bond types: overview and classification

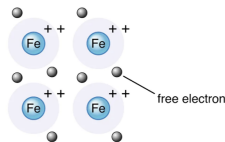
- ▶ **Ionic:** electron transfer creates oppositely charged ions (NaCl, MgO)
- ▶ **Metallic:** delocalized electron sea shared among positive ion cores (Cu, Al)
- ▶ **Covalent:** electron pairs shared between specific atoms (Si, diamond)
- ▶ **Van der Waals:** weak induced dipole attractions (noble gas solids)



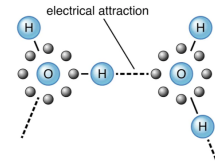
ionic bonding
electron transferred from Na to Cl



covalent bonding
atoms share electrons



metallic bonding
ions surrounded by free electrons



molecular bonding
weak electrical attraction binds molecules

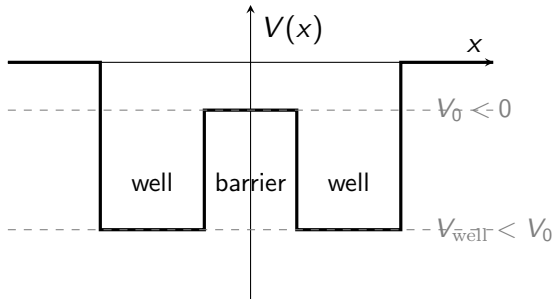
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Bonding types and properties

	Ionic	Covalent	Metallic	Secondary
Bond energy	Large	Variable	Variable	Smallest
Materials	Ceramics	Semiconductors, ceramics	Metals	Polymers
Melting Temp.	Large	Large (variable)	Moderate	Small
Stiffness	Large	Large (variable)	Moderate	Small
Thermal Exp.	Small	Small (variable)	Moderate	Large

Modeling electrons in bonds

- ▶ At the lowest energy levels, where do electrons prefer to stay? → at lower potentials.
- ▶ Is it favorable for electrons to stay above the barrier? → somewhat. Depends on the barrier height.
- ▶ See demonstration/visualization for a better insight on this

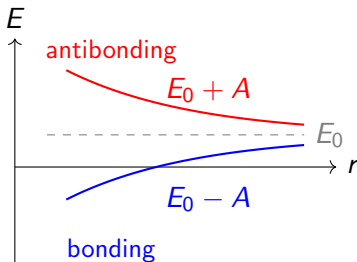


Energy splitting: bonding and antibonding states

If we simplify that problem:

$$E = E_0 \pm A \quad (5)$$

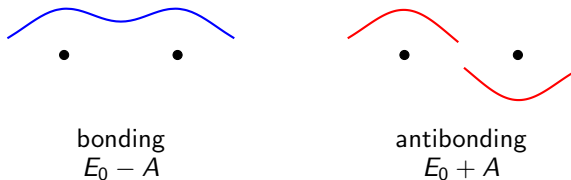
- ▶ **Lower state** $E_0 - A$: bonding (symmetric combination)
- ▶ **Upper state** $E_0 + A$: antibonding (antisymmetric combination)
- ▶ **Concept:** Bonding splits degenerate levels



H_2^+ energy levels and bonding

$$E_{\pm} = E_0 \pm A(R) \quad (6)$$

- ▶ **Bonding orbital** ($E_0 - A$): higher electron density between protons
- ▶ **Antibonding orbital** ($E_0 + A$): node between protons
- ▶ Electron occupies bonding orbital $\rightarrow$ stable molecule
- ▶ Bond length $\approx 1.06 \text{ \AA}$ where energy minimum occurs



Lecture 07b: free electron model; density of states

Prof. Daniel Schwalbe-Koda

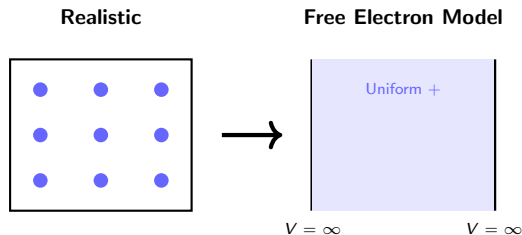
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The challenge: 10^{25} interacting electrons

- ▶ Metals contain $\sim 10^{25}$ valence electrons per $\text{cm}^3 \rightarrow$ exact many-body solution impossible
- ▶ Electrons interact via Coulomb repulsion and with $\sim 10^{23}$ ion cores
- ▶ Need drastic simplification that still captures essential physics

The free electron model: key assumptions

- ▶ Ignore electron-electron interactions (independent electron approximation)
- ▶ Replace ion lattice with uniform positive background charge (jellium model)
- ▶ Electrons are confined to a box with infinite potential walls at boundaries

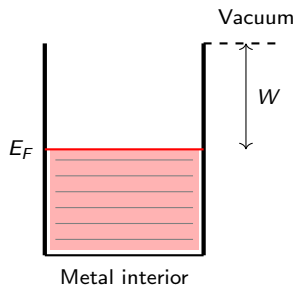


Why infinite potential well works

- ▶ Work function $\varphi \sim 4\text{--}5$ eV sets barrier height at metal surface
- ▶ Fermi energy $E_F \sim 5\text{--}10$ eV means most electrons deeply bound
- ▶ Infinite wall approximation valid:
 $\varphi \gg k_B T$ at room temperature

Thermal energy scale:

$$k_B T \approx 0.025 \text{ eV at } T = 300 \text{ K}$$



Particle in a 3D box: quantized energy

- ▶ Solve Schrödinger equation with $\psi = 0$ at walls (boundary conditions)
- ▶ Standing wave solutions require integer half-wavelengths fit in box
- ▶ Energy depends on three quantum numbers $n_x, n_y, n_z = 1, 2, 3, \dots$

Free electron energy in cubic box

$$E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

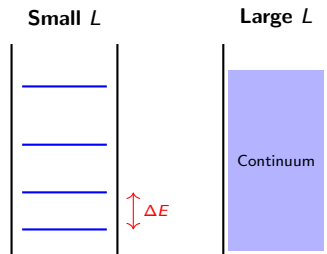
Notation: h = Planck constant, m = electron mass, L = box side length

Why this simple model captures metal physics

- ▶ Quantization explains discrete energy levels $\rightarrow$ basis for density of states
- ▶ Pauli exclusion fills states up to Fermi level $\rightarrow$ explains metallic properties
- ▶ Model predicts electrical conductivity, specific heat, emission phenomena
- ▶ **Limitations:** ignores band structure, electron-phonon coupling

Why energy becomes quasi-continuous

- ▶ Level spacing $\Delta E \sim h^2/(8mL^2)$ shrinks as L increases
- ▶ For $L = 1$ cm: $\Delta E \sim 10^{-15}$ eV — far smaller than $k_B T$
- ▶ Discrete quantum levels merge into effective continuum at macroscopic scales



Energy level spacing

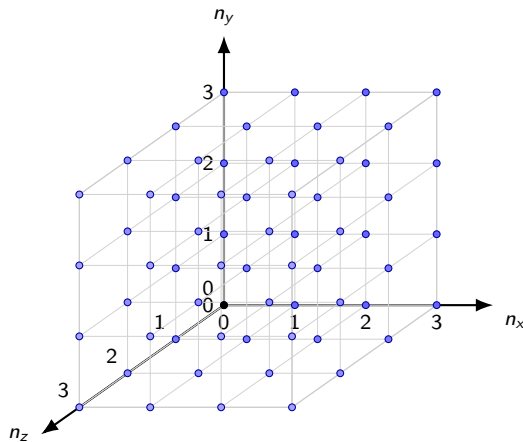
$$\Delta E \sim \frac{h^2}{8mL^2} \approx 10^{-15} \text{ eV for } L = 1 \text{ cm}$$

From discrete levels to counting states

- ▶ Energy quantized with positive integers n_x, n_y, n_z
- ▶ Each allowed state is a point in “ n -space” with coordinates (n_x, n_y, n_z)
- ▶ Goal: count states with energy $\leq E$ to find density of states $Z(E)$

Energy in terms of quantum numbers

$$E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

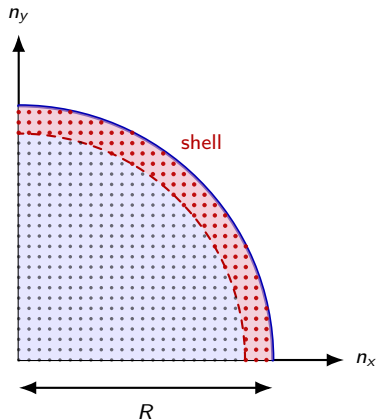


Constant energy surfaces in n-space

- ▶ Define $R^2 = n_x^2 + n_y^2 + n_z^2$, so $E = \frac{h^2}{8mL^2} R^2$
- ▶ States with energy $\leq E$ lie inside sphere of radius R
- ▶ Only positive octant counts: $n_x, n_y, n_z > 0$

Radius in n-space

$$R = \sqrt{\frac{8mL^2 E}{h^2}}$$



Counting states: volume in n-space

- ▶ Total states with energy $\leq E$:

$$N(E) = \frac{1}{8} \times \frac{4\pi}{3} R^3 \times 2 \quad (\text{spin factor})$$

- ▶ Substitute $R = \sqrt{8mL^2E/h^2}$
- ▶ Density of states $Z(E) = dN/dE$ gives states per unit energy

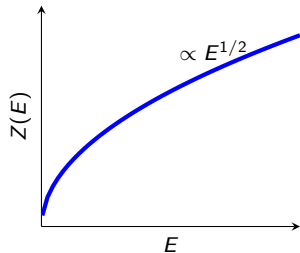
Integrated density of states

$$N(E) = \frac{\pi}{3} \left(\frac{8mL^2E}{h^2} \right)^{3/2}$$

Note: Factor of 2 accounts for spin-up and spin-down electrons (Pauli principle)

Density of states $Z(E) \propto E^{1/2}$

- ▶ Differentiate $N(E)$: $Z(E) = \frac{dN}{dE} = C \cdot E^{1/2}$
- ▶ Constant $C = \frac{4\pi L^3(2m)^{3/2}}{h^3}$ depends on volume and mass
- ▶ Square-root dependence is signature of 3D free electrons



Density of states for free electrons

$$Z(E) = \frac{4\pi L^3(2m)^{3/2}}{h^3} E^{1/2}$$

Checkpoint: density of states derivation

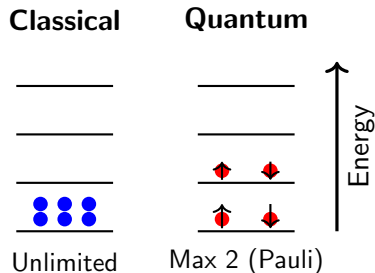
Test your understanding:

1. Why do we only count the positive octant in n -space?
▷ *Hint: What are the allowed values of n_x, n_y, n_z ?*
2. Where does the factor of 2 for spin enter the derivation?
▷ *Hint: How many electrons can occupy each spatial state?*
3. Explain physically why $Z(E) \propto E^{1/2}$ in 3D but not in 1D or 2D
▷ *Hint: How does the “surface area” in n -space scale with dimension?*

Why classical statistics fail for electrons

The problem:

- ▶ **Classical Maxwell-Boltzmann:** any particle can occupy any energy state
- ▶ **Electrons are fermions:** Pauli exclusion forbids identical quantum states
- ▶ At metal densities ($\sim 10^{28}/\text{m}^3$), quantum effects dominate even at room T



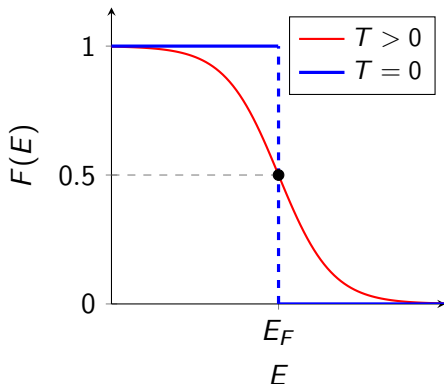
The Fermi-Dirac distribution function

Key properties:

- ▶ $F(E)$ = probability state at energy E is occupied
- ▶ At $E = E_F$: occupation probability is exactly $\frac{1}{2}$
- ▶ Interpolates between fully occupied ($E \ll E_F$) and empty ($E \gg E_F$)

Fermi-Dirac distribution

$$F(E) = \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right) + 1}$$



Fermi-Dirac at $T = 0$: the sharp cutoff

Zero temperature limit:

- ▶ At $T = 0$: all states below E_F are filled, all above are empty
- ▶ $F(E)$ becomes a step function
- ▶ Electrons fill lowest available states like water filling a container

$T = 0$ limit

$$F(E)|_{T=0} = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$$

Physical picture: Pauli exclusion forces electrons to stack up in energy, filling states from bottom up until all N electrons are accommodated.

Calculating E_F : counting electrons

Strategy: Total electrons = integral of occupied states

- ▶ At $T = 0$: integrate $Z(E)$ from 0 to E_F since $F(E) = 1$ below Fermi level
- ▶ Substitute $Z(E) = C \cdot E^{1/2}$ and solve for E_F

Total number of electrons N

$$\begin{aligned} N &= \int_0^{E_F} Z(E) dE = C \int_0^{E_F} E^{1/2} dE \\ &= C \cdot \frac{E^{3/2}}{3/2} \Big|_0^{E_F} = \frac{2C}{3} E_F^{3/2} \end{aligned} \quad (1)$$

Notation: $C = \frac{4\pi V(2m)^{3/2}}{h^3}$ includes factor of 2 for spin

Fermi level depends only on electron density

Solving for E_F :

- ▶ Invert $N = \frac{2C}{3} E_F^{3/2}$ to find E_F
- ▶ Result depends on N/V (electron density) only
- ▶ No temperature dependence at $T = 0$; weak T -dependence at finite T

Fermi Level

$$E_F = \frac{\hbar^2}{2m} \left(3\pi^2 \frac{N}{V} \right)^{2/3}$$

Typical values for metals:

- ▶ $E_F \sim 5\text{--}10$ eV
- ▶ Fermi temperature: $T_F = E_F/k_B \sim 50,000$ K $\gg$ room temperature

Checkpoint: Fermi-Dirac statistics

Test your understanding:

Q1: Why is $F(E_F) = \frac{1}{2}$ regardless of temperature?

▷ *Hint: Substitute $E = E_F$ into the Fermi-Dirac formula*

Q2: At $T = 0$, a metal has N electrons. Where do they reside in energy space?

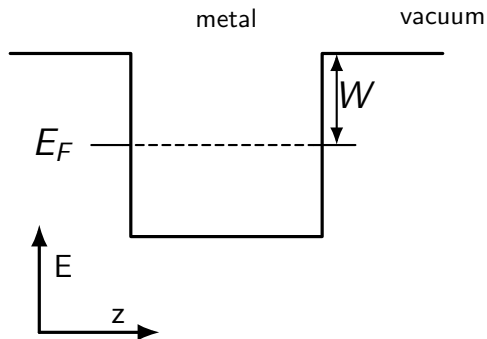
▷ *Hint: What does the step function tell you?*

Q3: If electron density doubles, by what factor does E_F change?

▷ *Hint: $E_F \propto (N/V)^{2/3}$*

Work function and thermionic emission

- ▶ Electrons at high T gain enough thermal energy to escape metal surface
- ▶ Only electrons with $E > E_F + W$ can overcome work function barrier
- ▶ Emission current depends on **tail** of Fermi-Dirac distribution above barrier
- ▶ Circuit must replenish emitted electrons for steady current



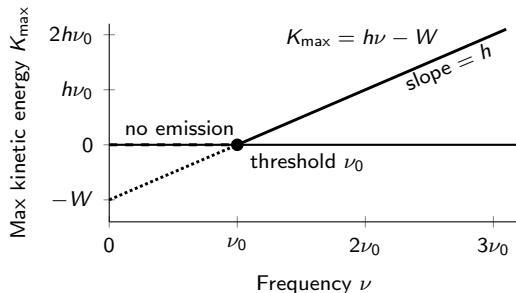
Photoelectric effect

Classical wave theory predicts:

- ▶ Any frequency works if intensity is high enough
- ▶ Energy accumulates continuously from wave

Experiment shows:

- ▶ Threshold frequency ν_0 exists regardless of intensity
- ▶ Electron KE depends on frequency, *not* intensity



Wave picture cannot explain threshold or frequency dependence!

Einstein's explanation of the photoelectric effect

Key insight: Light consists of *photons* with quantized energy $E = h\nu = \hbar\omega$

Photoelectric threshold

$$\hbar\omega > W \quad \Rightarrow \quad KE_{\max} = \hbar\omega - W$$

Physical picture:

- ▶ One photon (energy $E = \hbar\omega$) ejects one electron
- ▶ Threshold: photon energy must exceed work function W
- ▶ $KE_{\max}$ linear in frequency (explains experimental data exactly)

Notation:

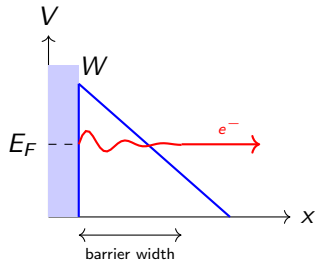
- ▶ $\hbar\omega$ = photon energy, W = work function
- ▶ Threshold frequency: $\nu_0 = W/h$

Field emission: cold cathode tunneling

- ▶ At very high fields ($\sim 10^9$ V/m), electrons **tunnel** through barrier
- ▶ **Temperature-independent:** quantum tunneling replaces thermal activation
- ▶ Barrier width decreases with increasing field $\Rightarrow$ exponentially increasing tunneling

Applications:

- ▶ Field-emission microscopy
- ▶ Cold cathode displays
- ▶ Sharp tip electron sources



Contact potential: Fermi level equalization

Two metals with different work functions:

- ▶ $W_1 < W_2$ means $E_{F,1} > E_{F,2}$ (relative to vacuum)
- ▶ Electrons flow from metal 1 $\rightarrow$ metal 2
- ▶ Equilibrium: Fermi levels align

Contact potential difference:

$$V_{\text{contact}} = \frac{W_2 - W_1}{e}$$

